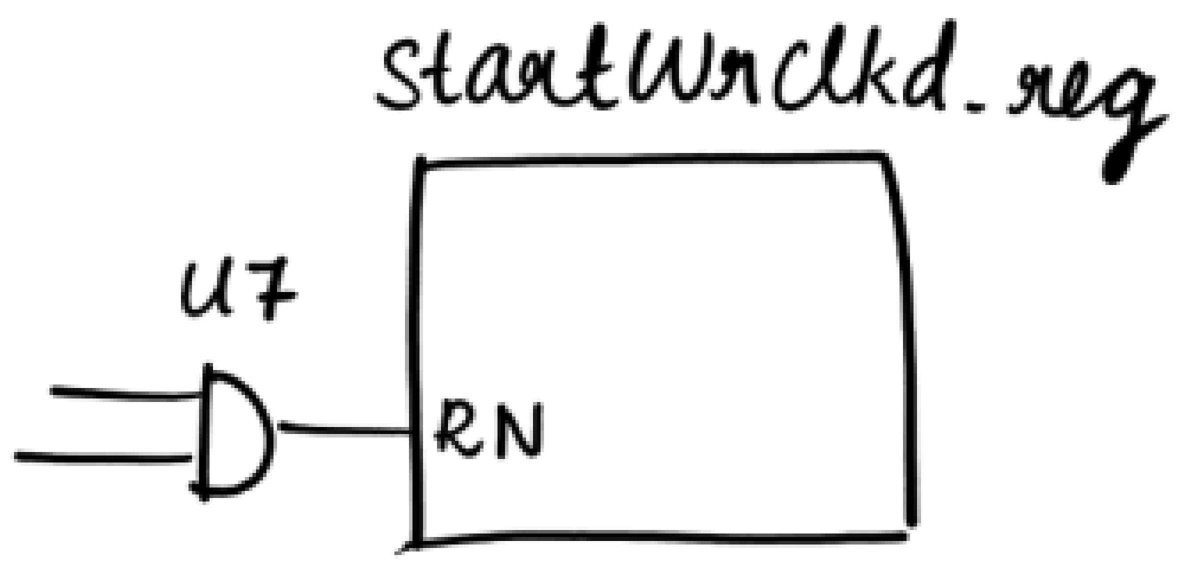
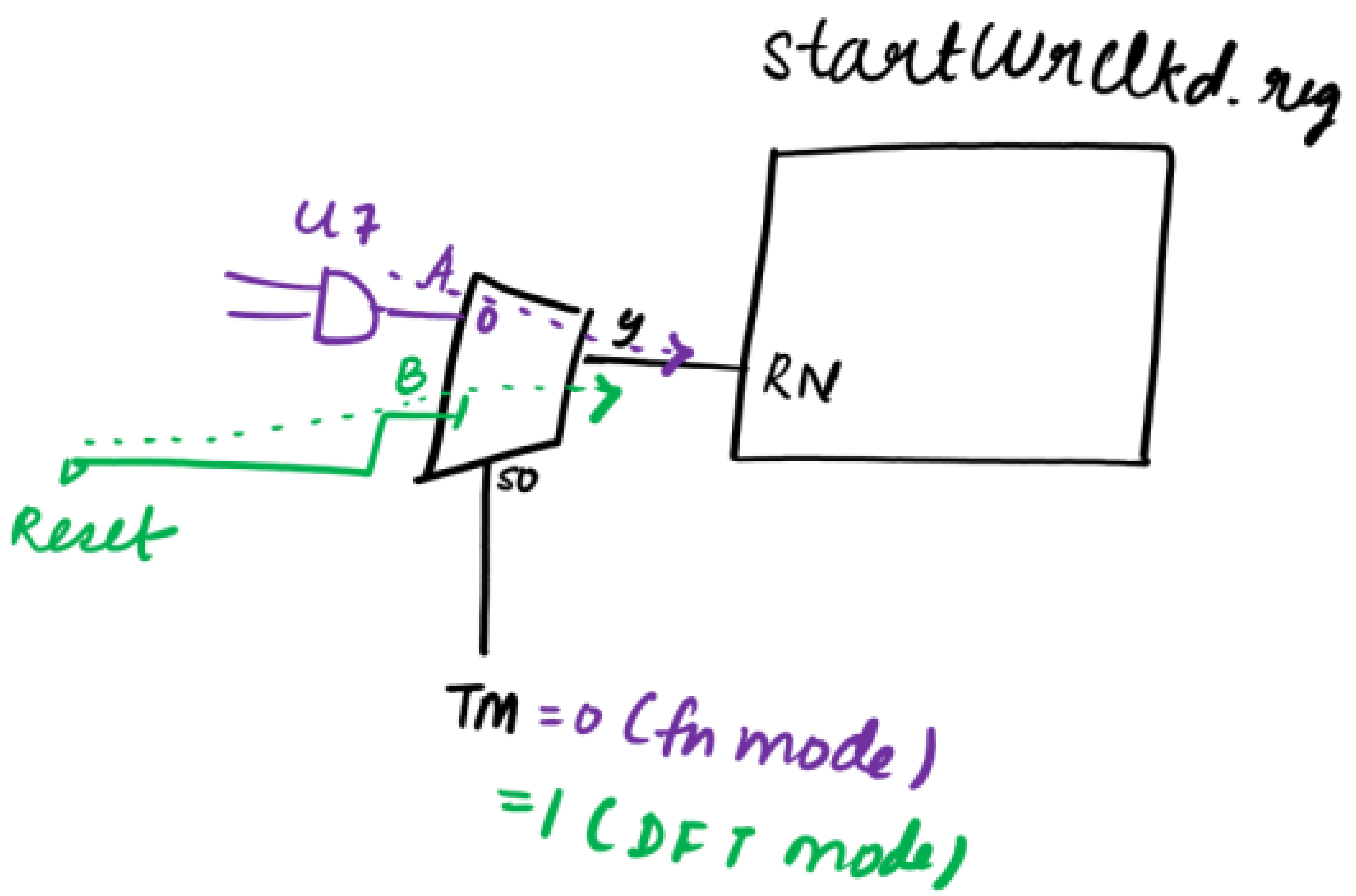
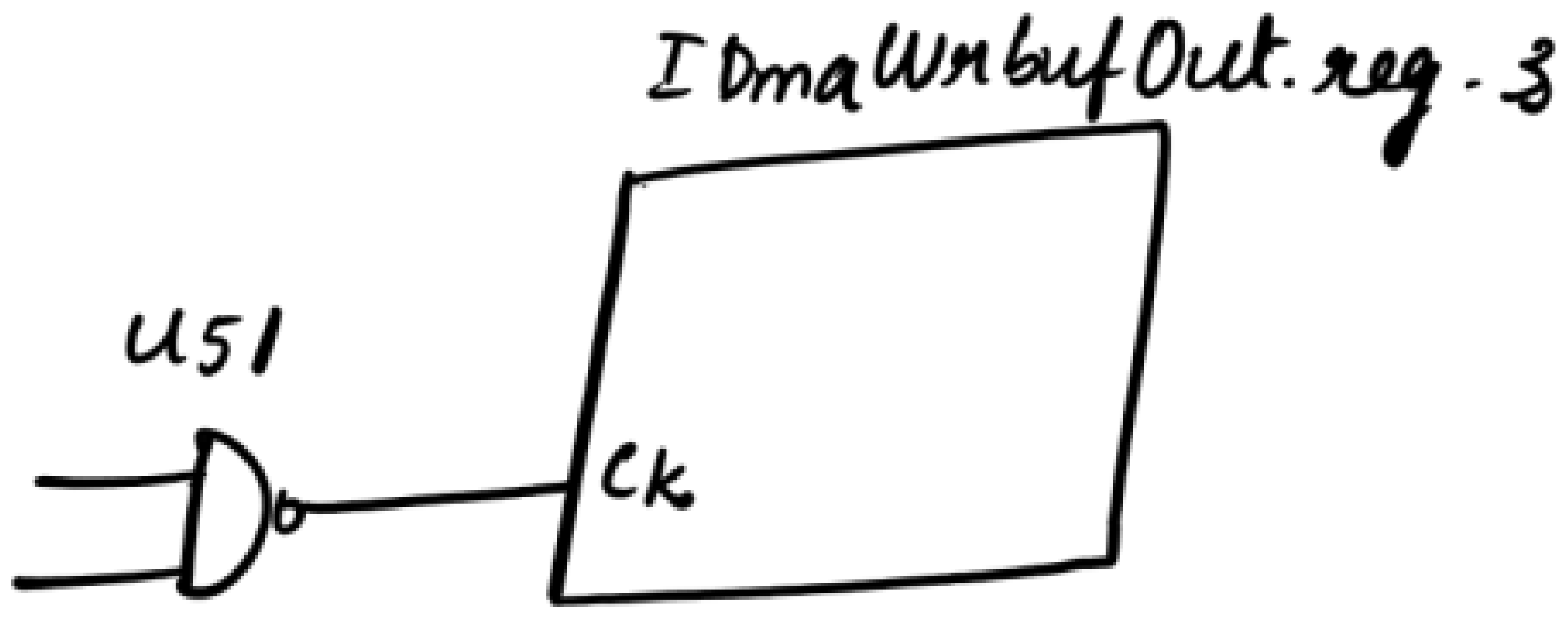
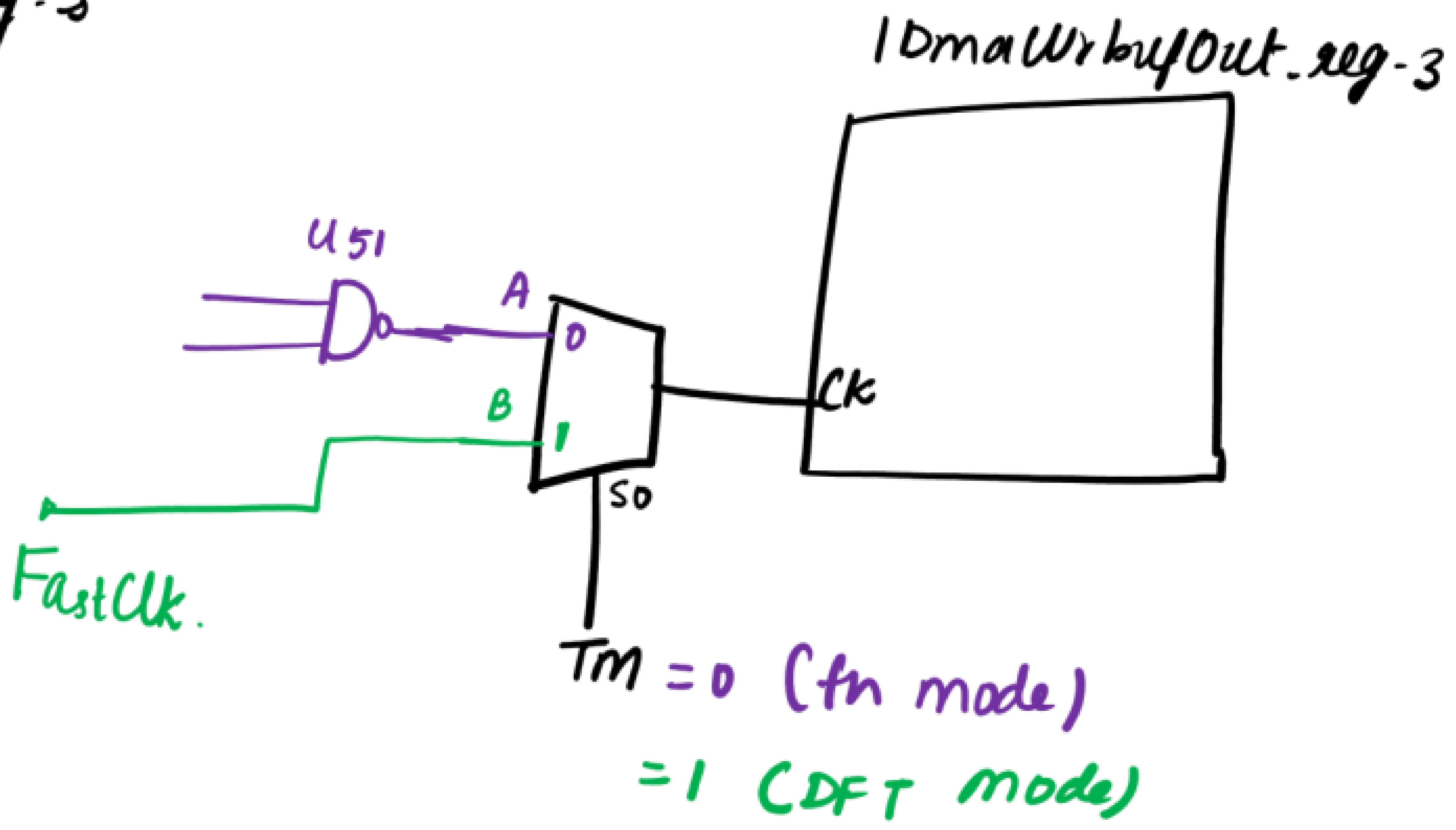
CASE 2 :-

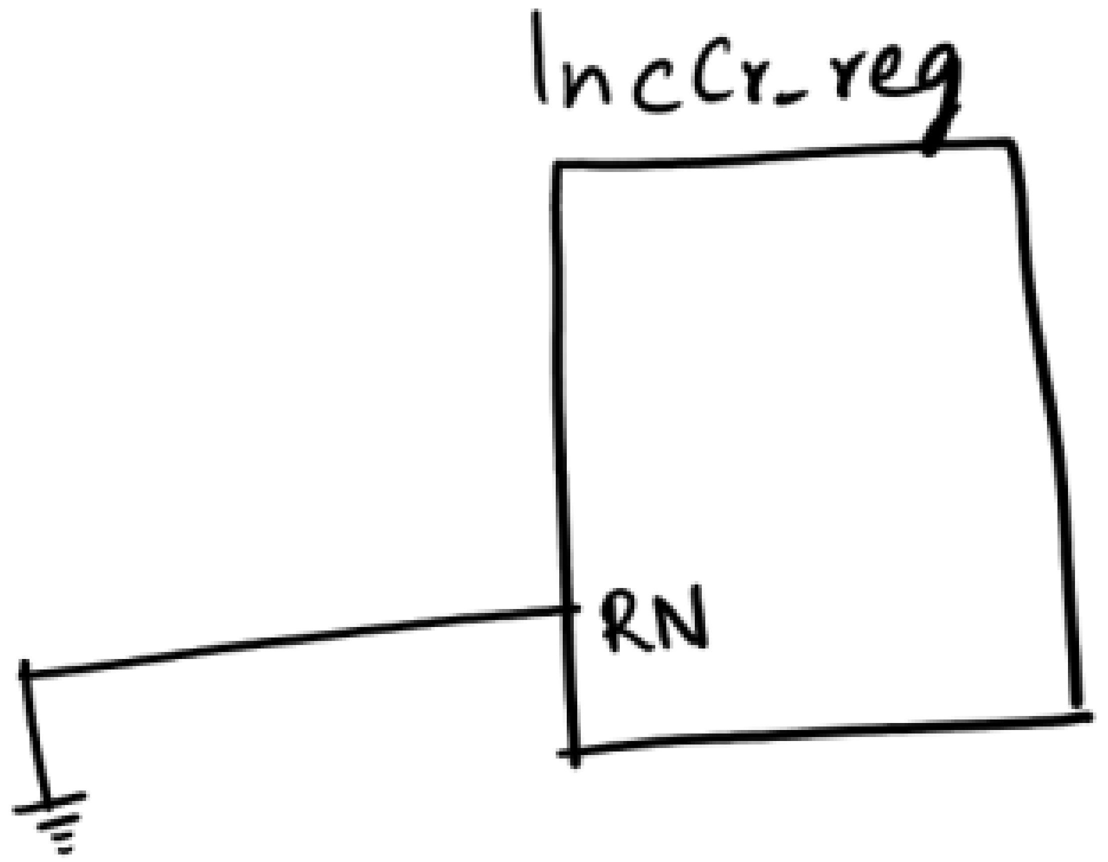
SI VIOLATION :-

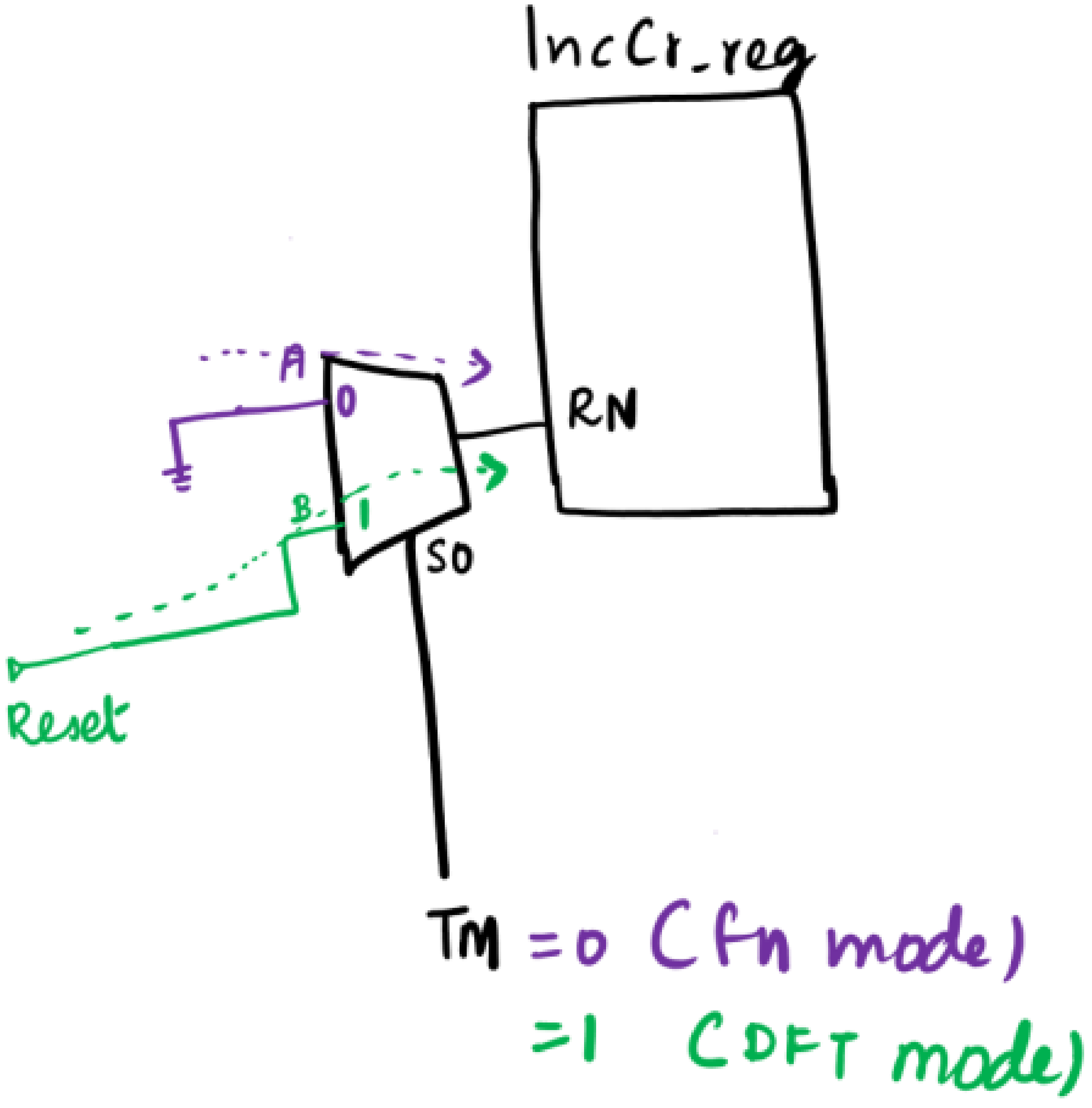


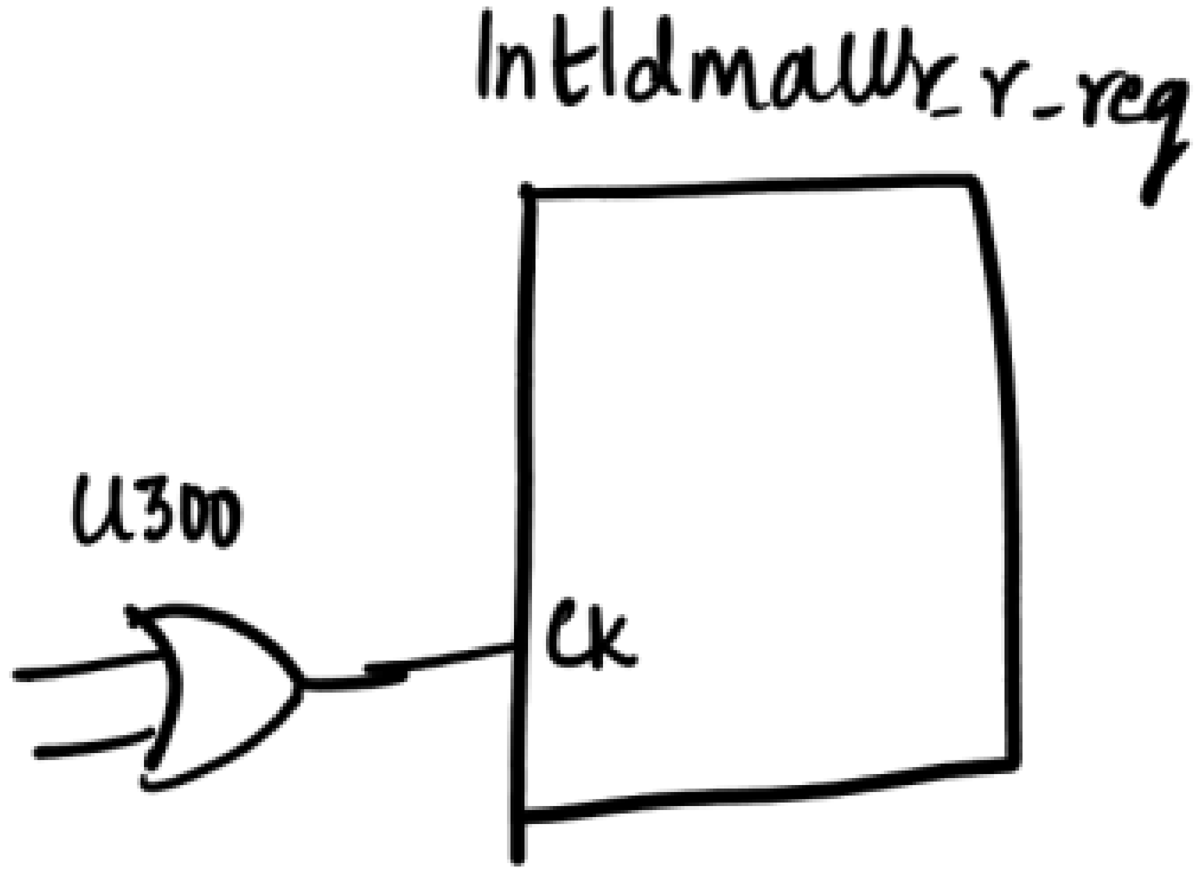


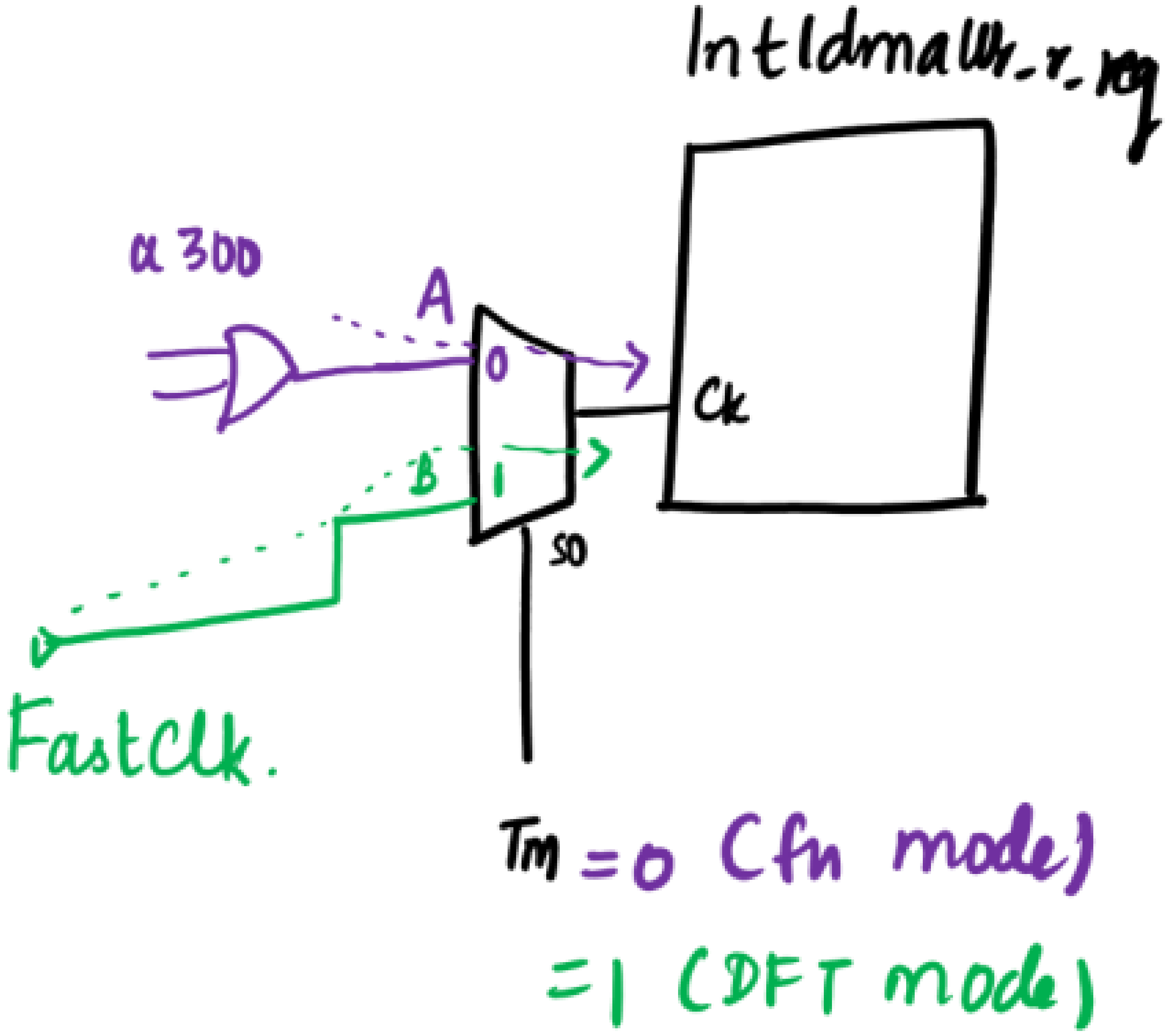


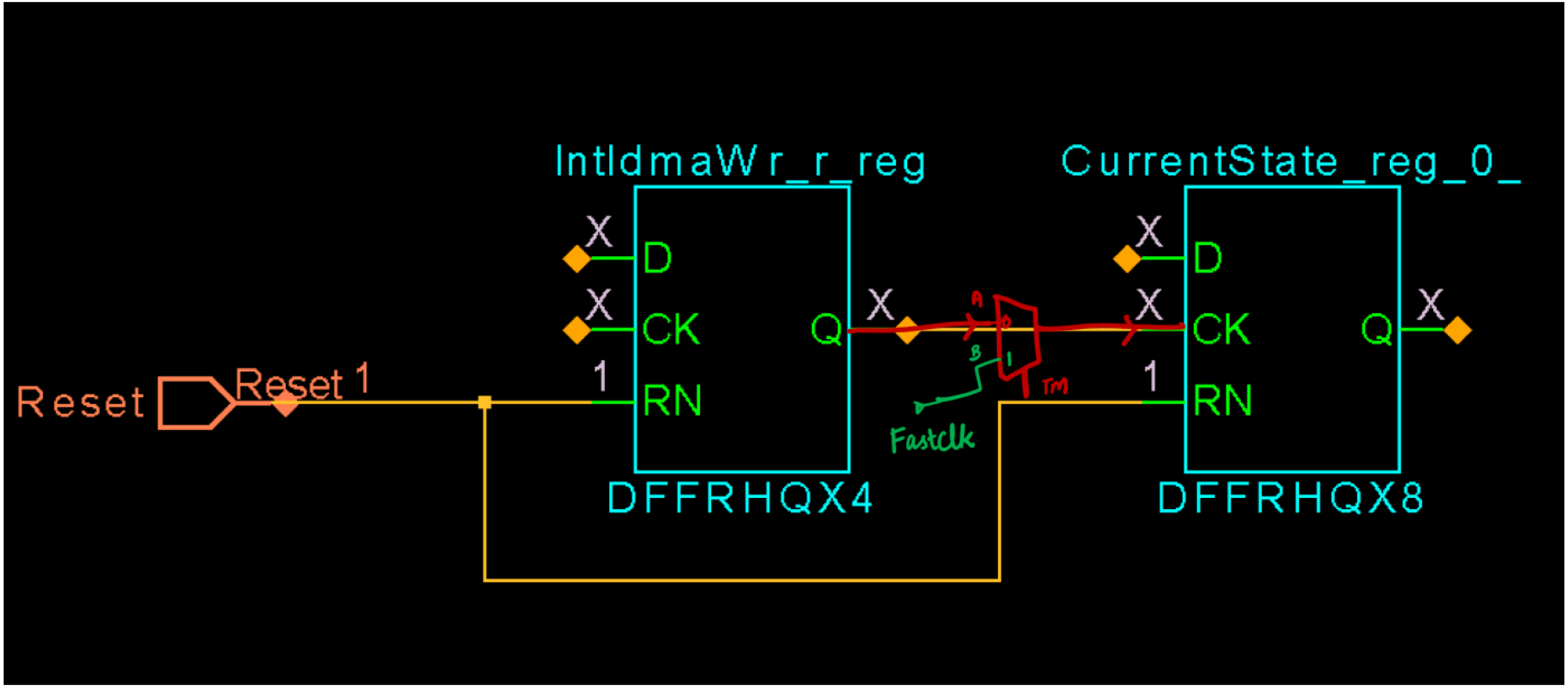


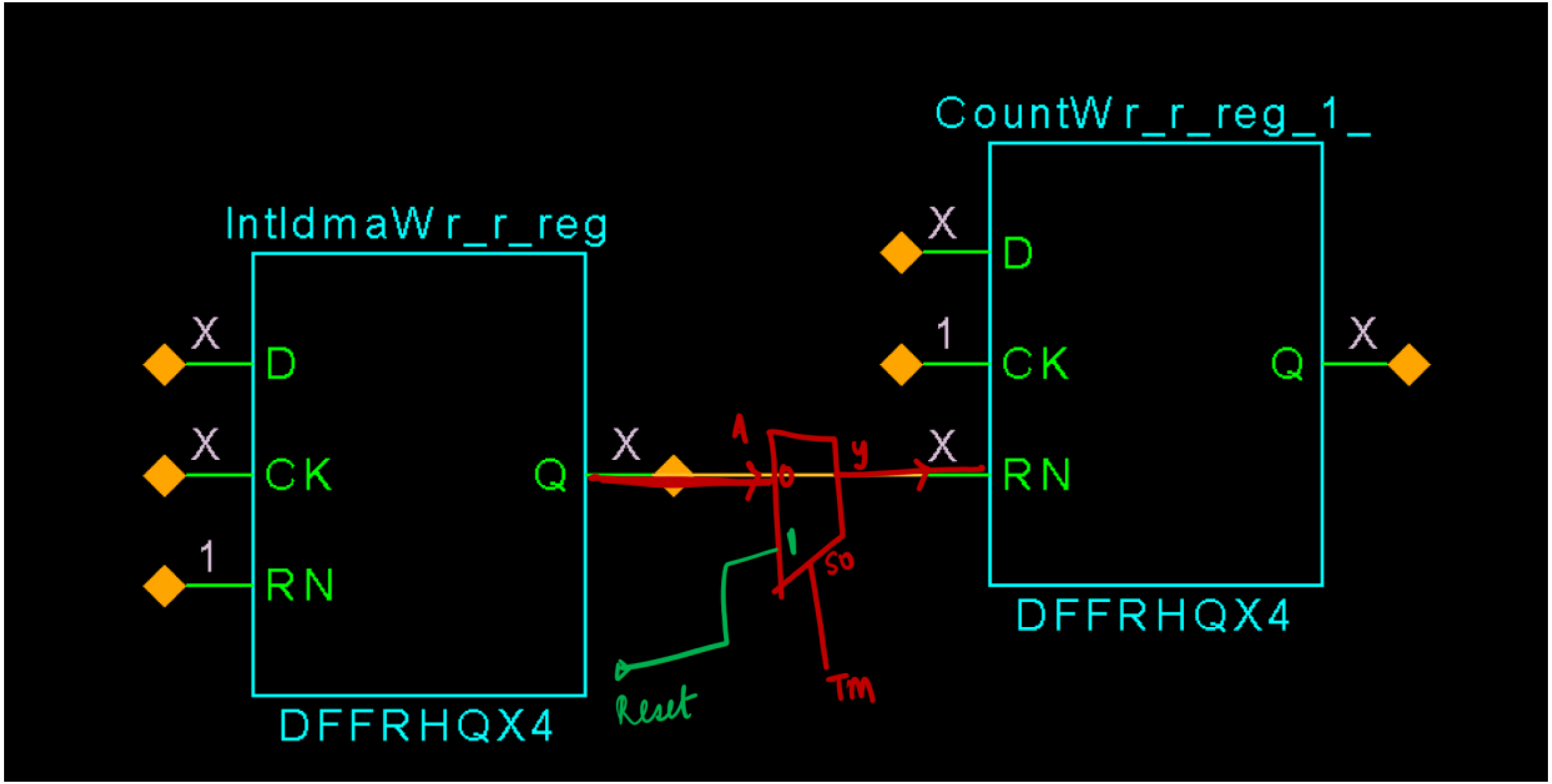

IncCr_reg
RN


IncCr_reg
A
0
B
1
RN
SO
Reset
TM =0 (fn mode)
=1 (DFT mode)

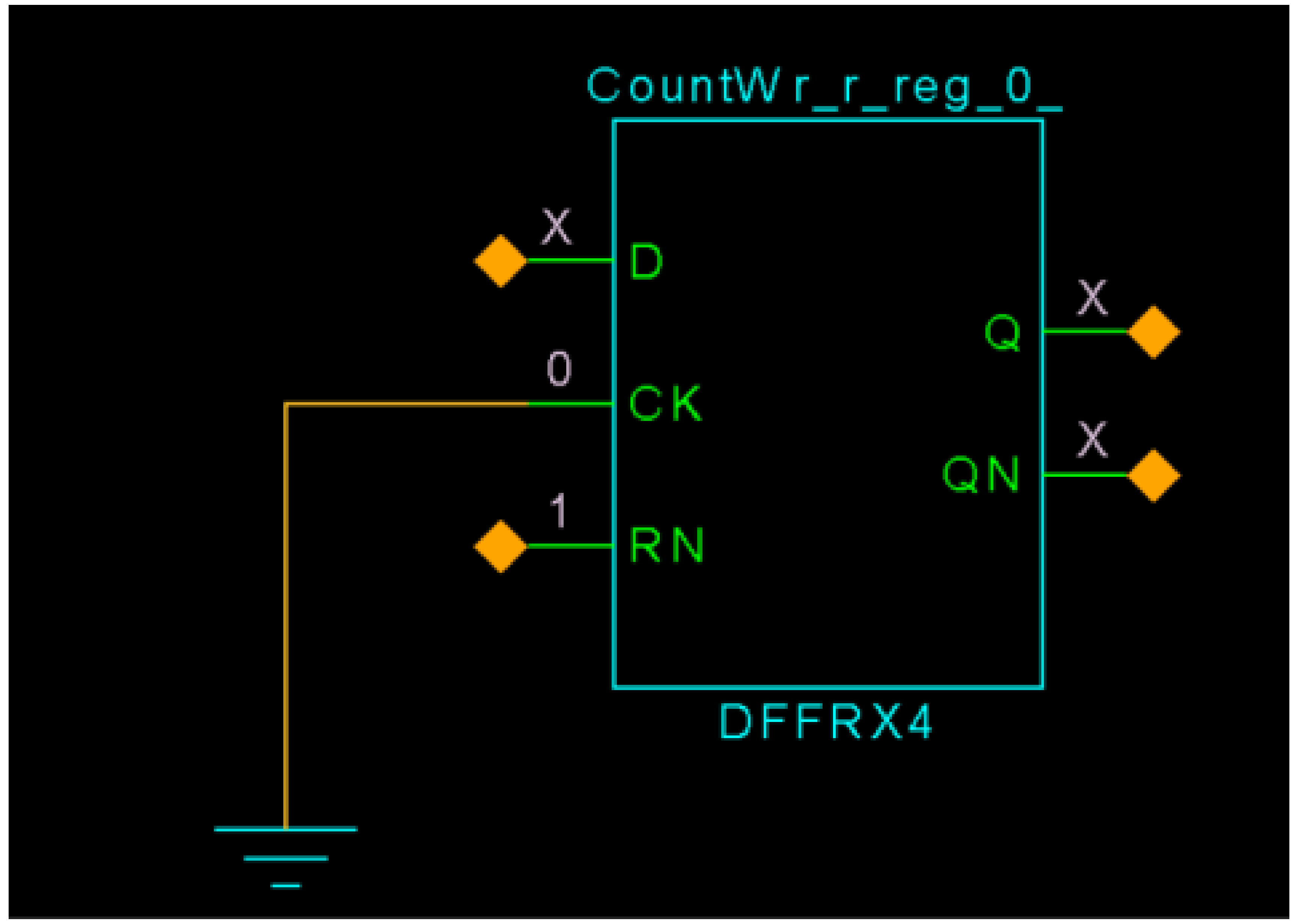

Intldmawr_r_reg
U300
Ck

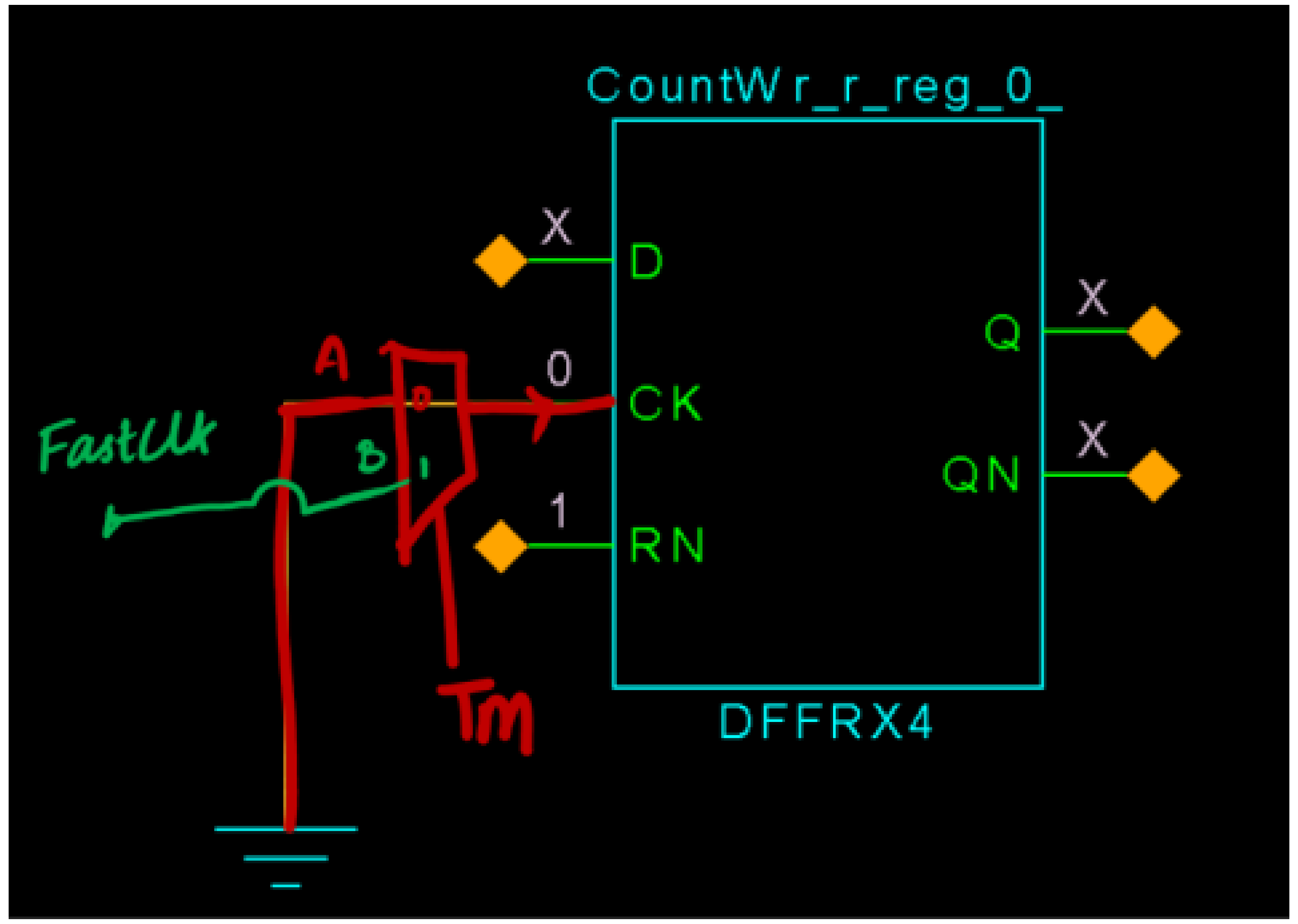

Intldmawr_r_reg
U300
A
0
B
1
Ck
SO
FastClk.
TM =0 (fn mode)
=1 (DFT mode)

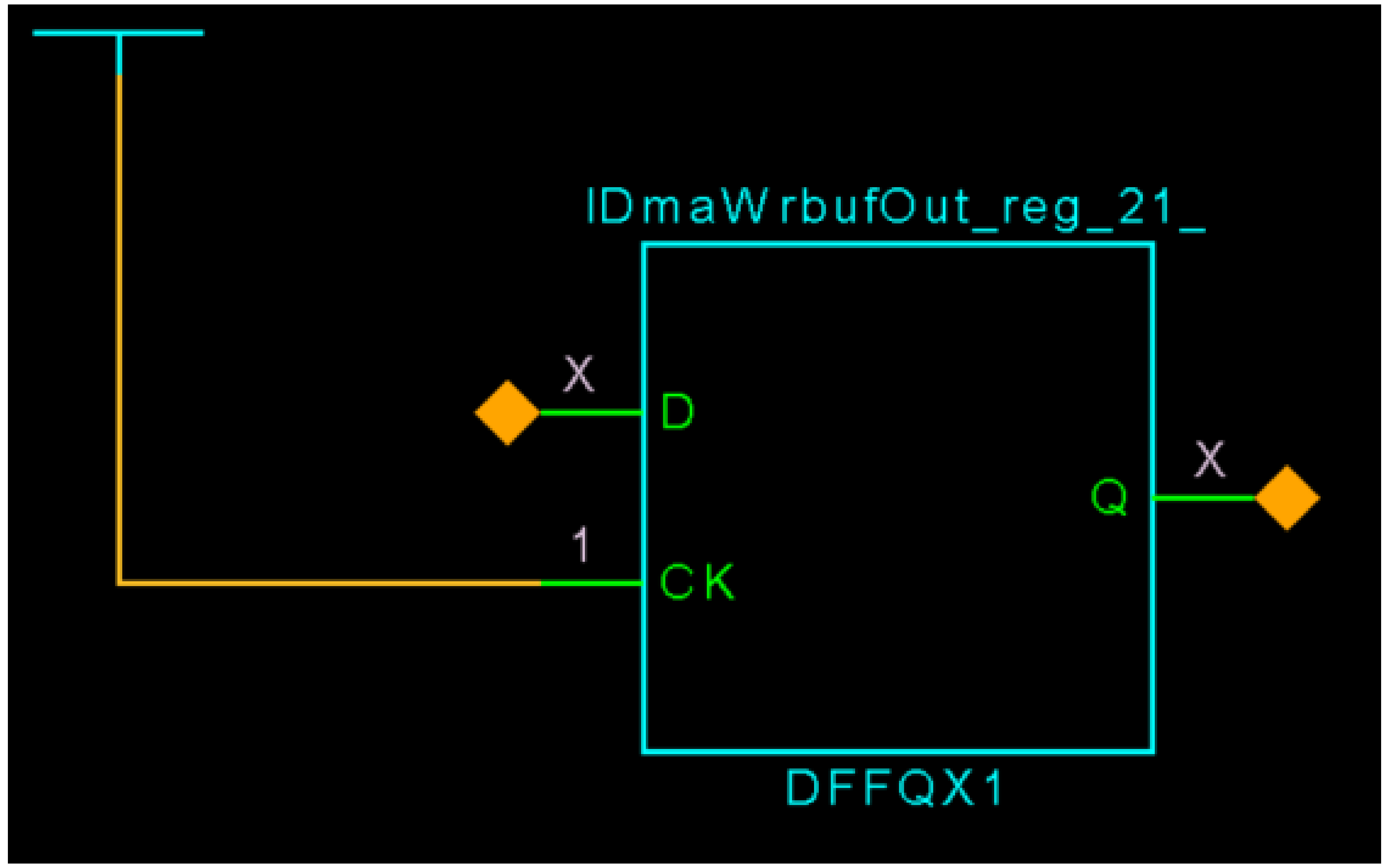

IntldmaWr_r_reg
CurrentState_reg_0_
D
CK
Q
RN
X
1
Reset
Reset 1
DFFRHQX4
DFFRHQX8
A
B
0
1
TM
Fastclk

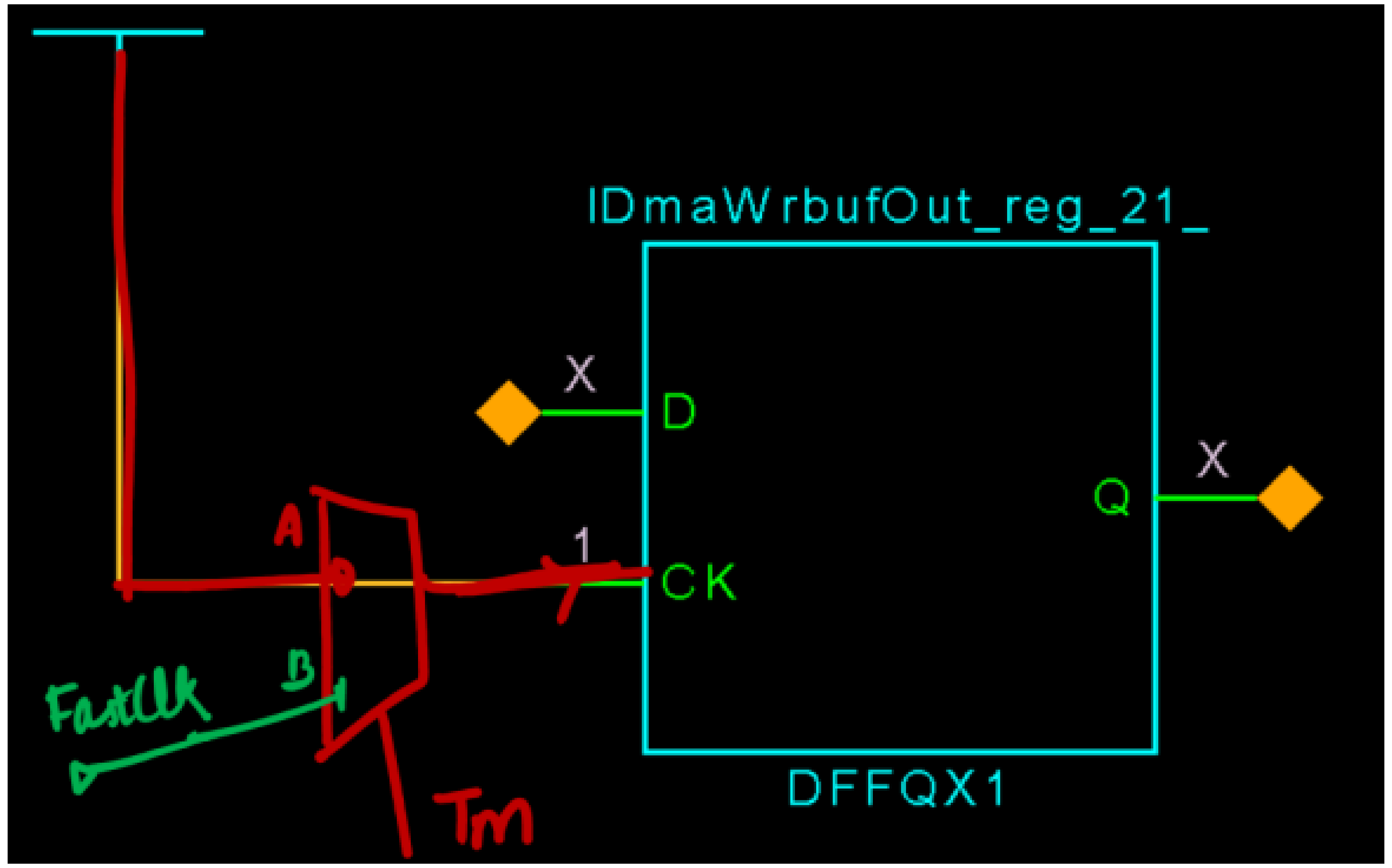

IntldmaWr_r_reg
CountWr_r_reg_1_
X
D
X
CK
1
RN
Q
X
DFFRHQX4
X
D
1
CK
X
RN
Q
X
DFFRHQX4
A
y
0
1
S0
Reset
TM

S2 VIOLATION :-

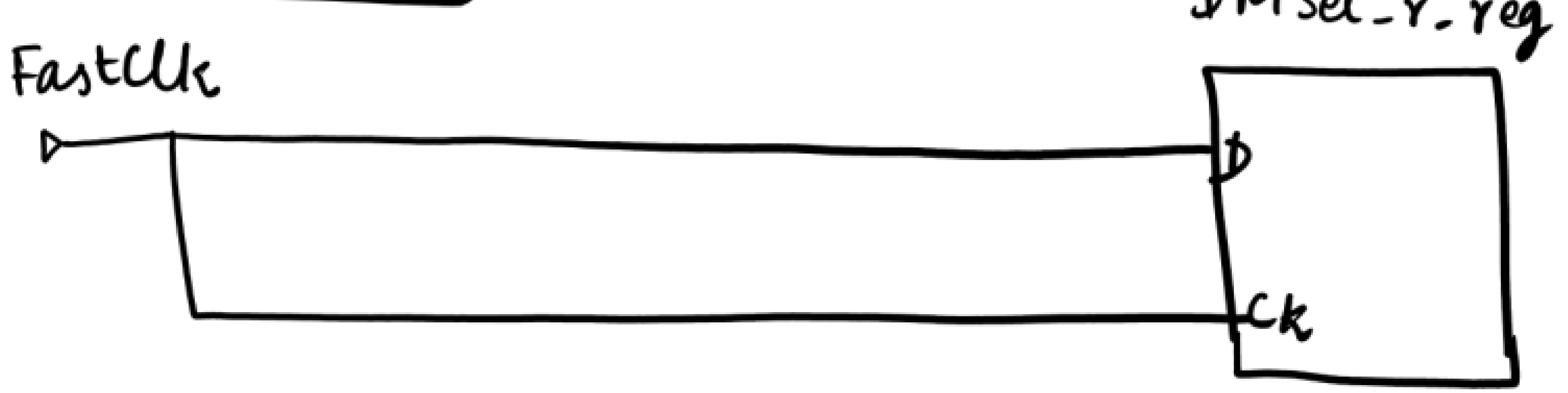


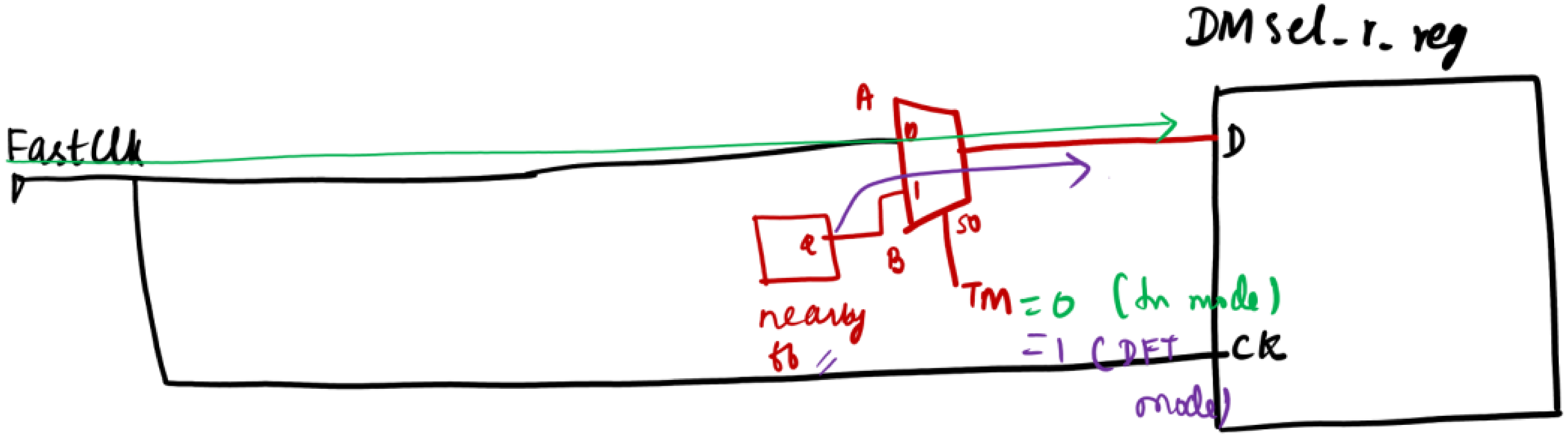


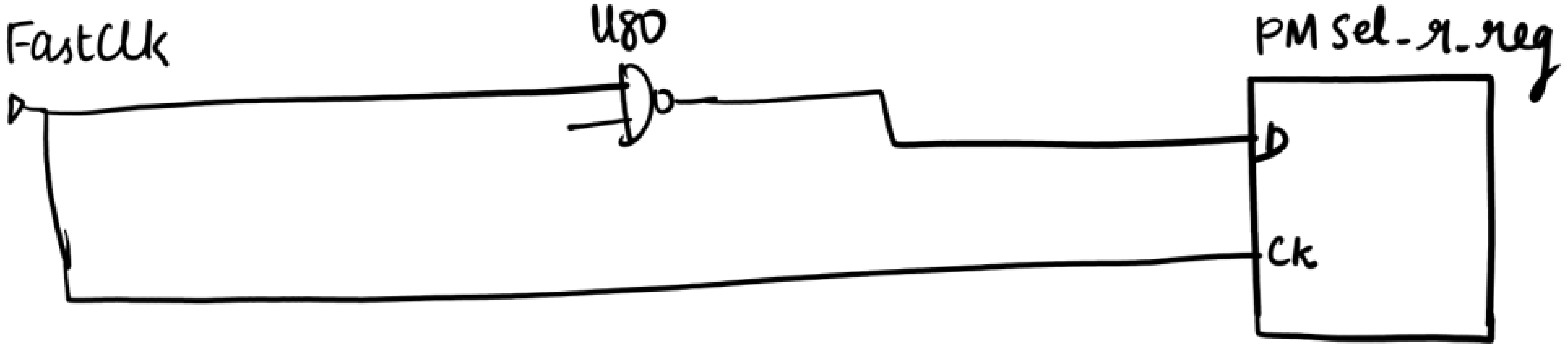


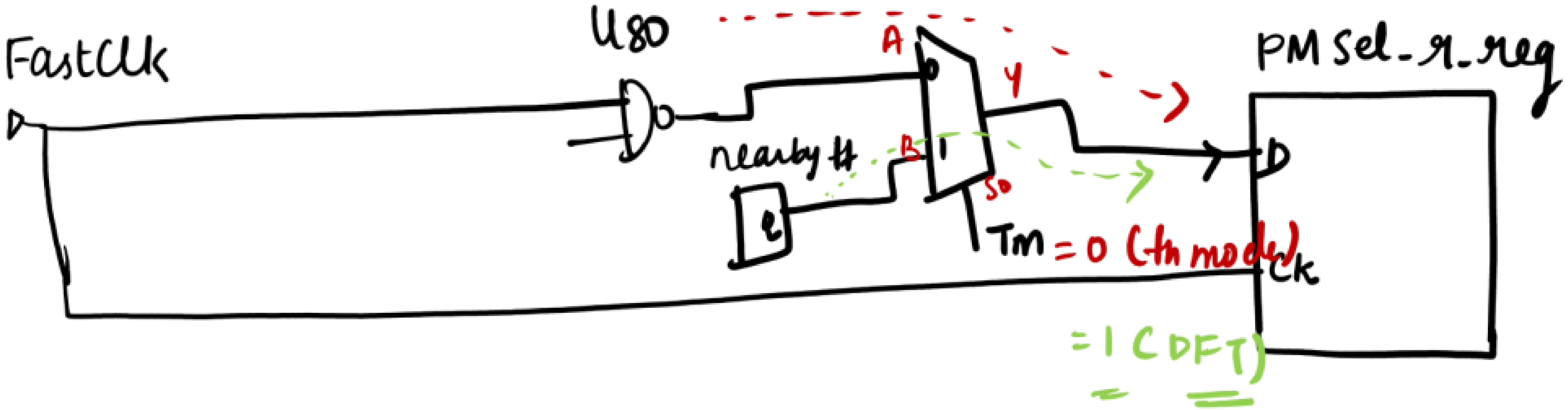


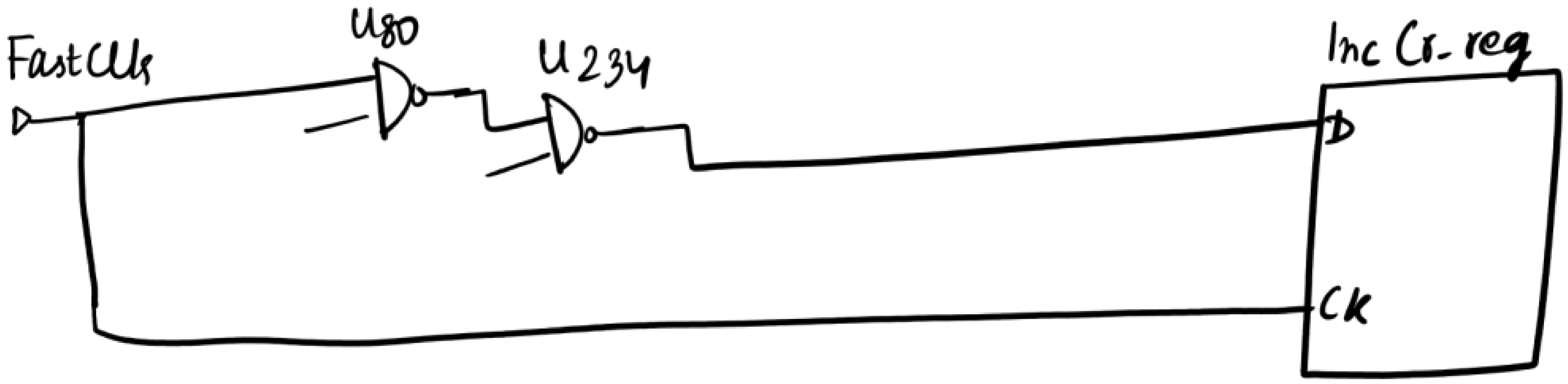
C6 VIOLATION :-



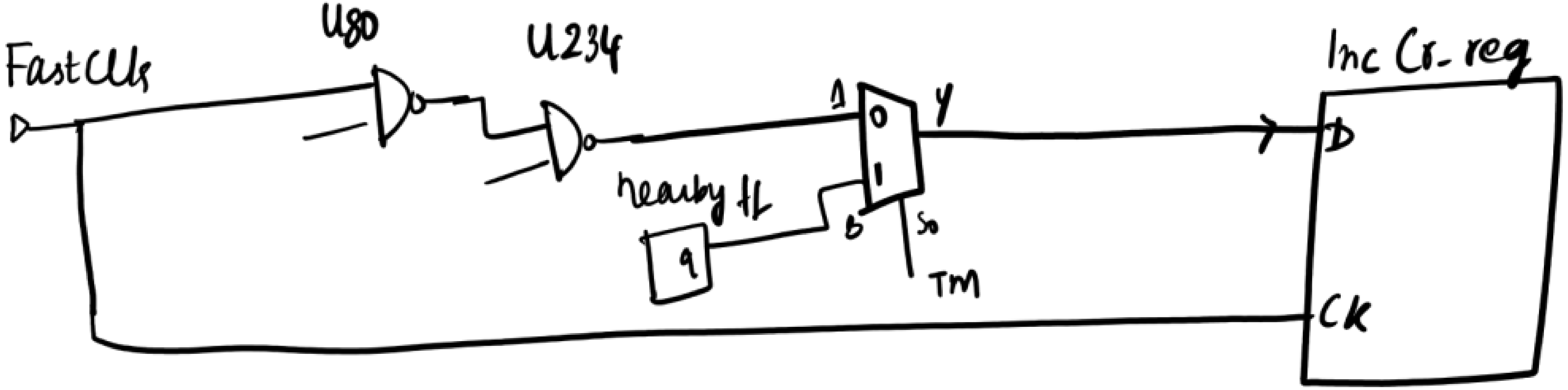



FastClk
U80
PMSel_r_reg
Ck


FastClk
U80
A
Y
B
1
S0
nearby ff
PMSel_r_reg
TM = 0 (fn mode)
Ck
= 1 (DFT)



FastClk
U80
U234
Inc Cr_reg
D
CK

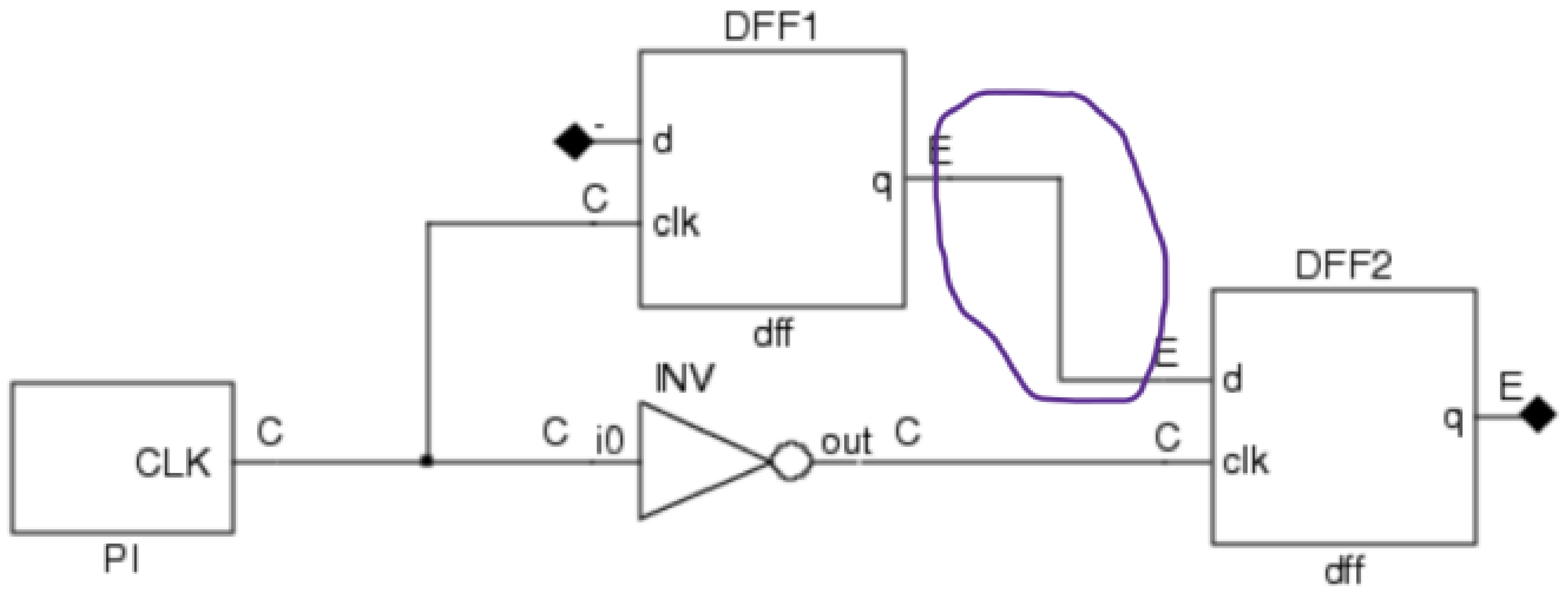


FastClk
U80
U234
A
0
1
Y
B
S0
TM
nearby ff
q
Inc Cr_reg
D
CK

DRCs (Contd ...)

C3 :

Figure 1. C3 Violation Example



Note:

The same situation would occur if the clock pin was connected directly (rather than through an inverter) to DFF2, and DFF2 was negative-edge triggered.

Whenever we have both positive and negative edge triggered flops, for scan path, it will taken care as discussed in theory class :

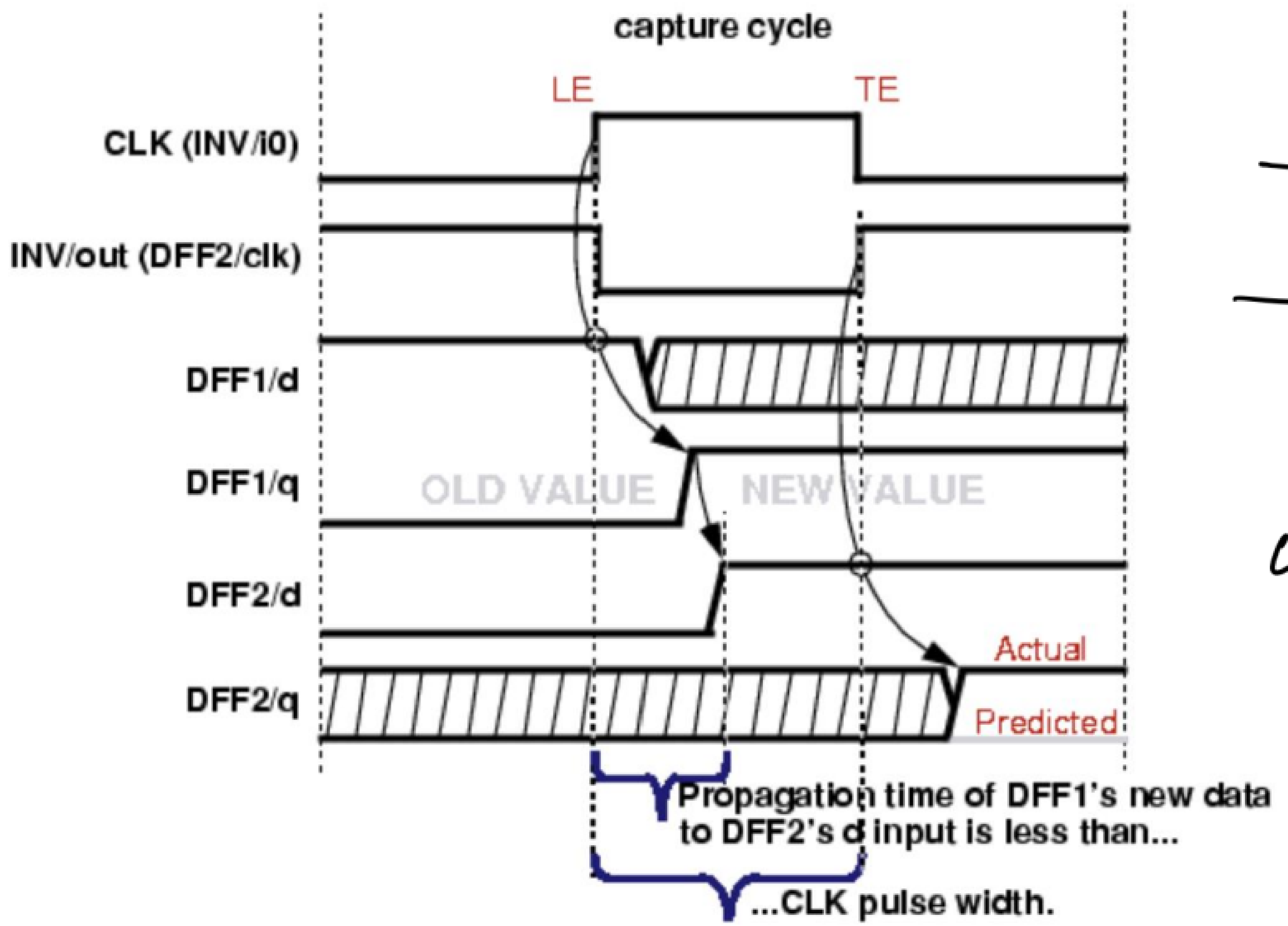
- While doing scan chain stitching, the tool will try to keep all negative flops at the starting of the scan chain.
- If suppose the tool is not able to do so, then if there is positive ff and then negative ff, then the tool will add LL.

For C3, the problem is happening at the D path (marked in violet in the fig). This can cause problem during capture phase. Failure to address this issue can result in simulation mismatches when we verify the test patterns in a timing-based simulator.

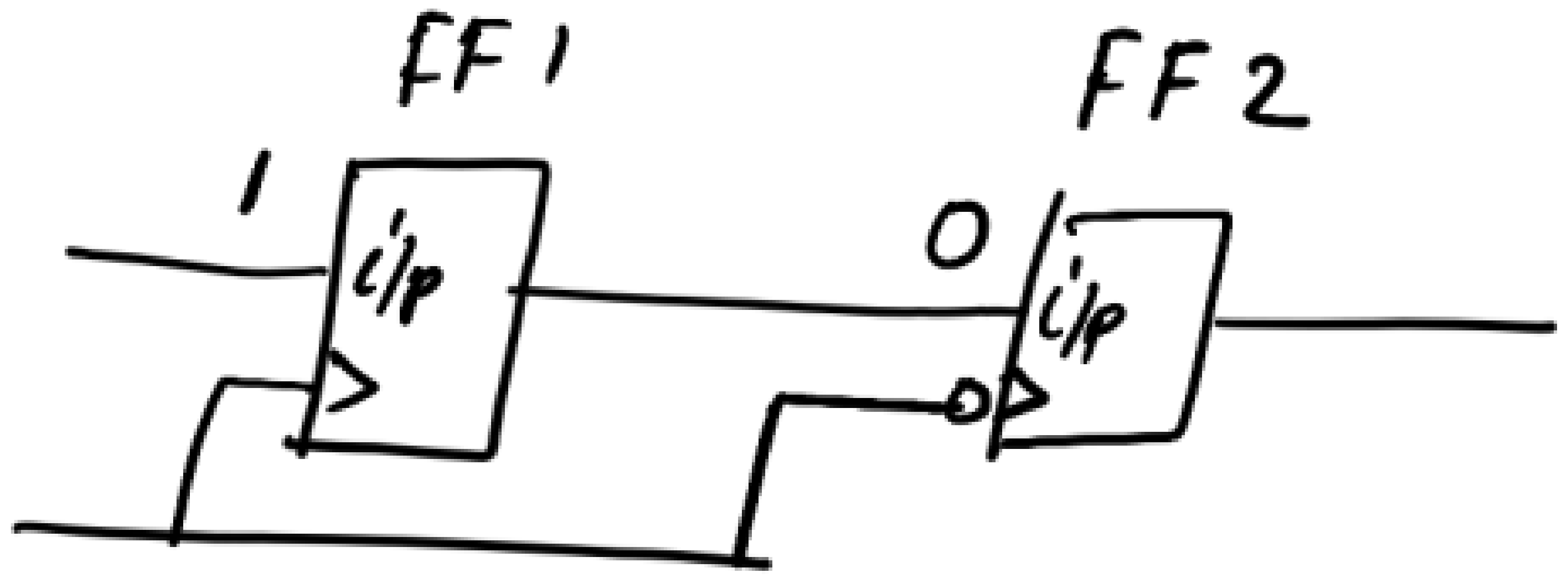
Command : set_split_capture_cycle on → Given during ATPG

When set to on, the tool updates simulation data between clock edges. This “split capture cycling” of data allows the tool to determine correct capture values for trailing edge and level-sensitive state elements, even in the presence of C3 violation.

When set to “off”, simulation data is updated once each clock cycle.

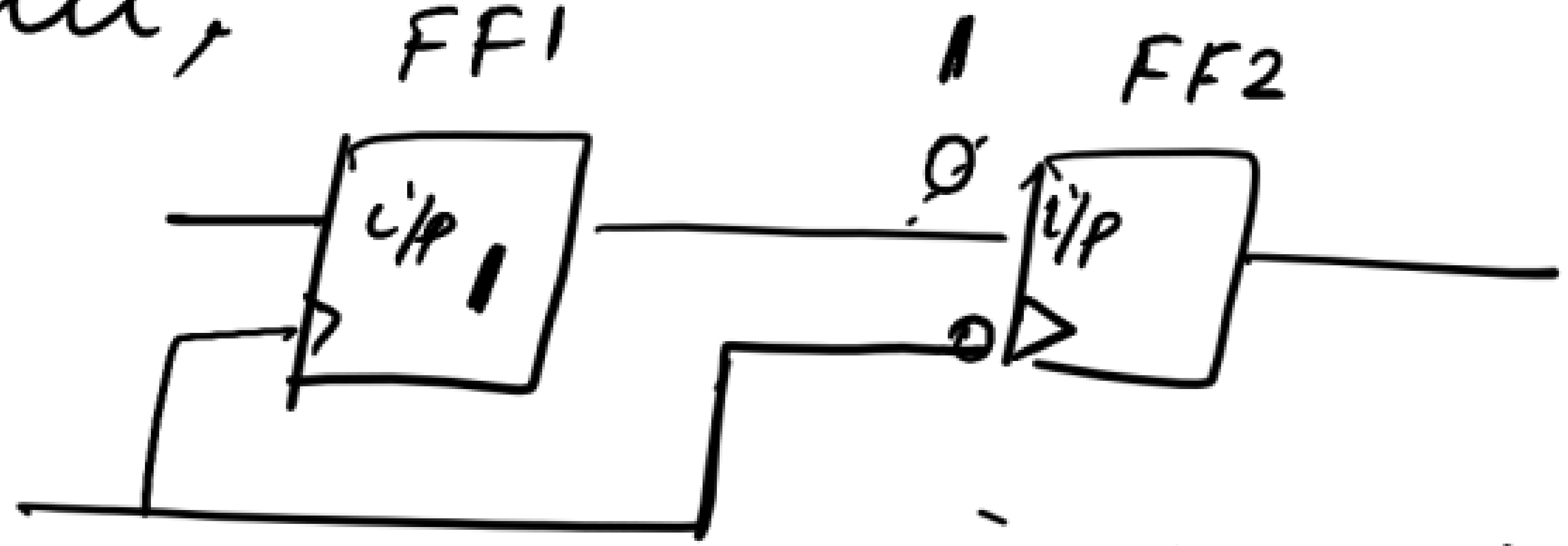


Example Timing That Allows Sink to Capture Source's New Data



ATPG tool will predict that FF2 will capture 0 (the value there at input of FF2 before clk pulse)

But,



the value captured by F1 during the rising edge of clock will overwrite the value at the i/p of FF2 before the falling edge of clk. So FF2 will capture 1.

NOTE :-

You can enable the '-edge merge' & '-clock merge' switch in the insert_test_logic command for all the designs.

If there are both +ve and -ve ET flops in the design, and if there are multiple scan clks, then the tool will go for edge mixing and domain mixing as we have enabled these 2 switches.

Even if you don't have both +ve and -ve E.T ffs and multiple scan clks in the design, enabling those 2 switches doesn't cause any harm.